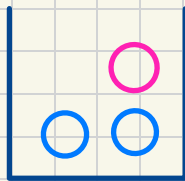


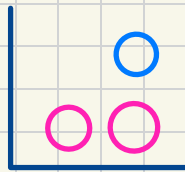
Classification.

Inspiration.

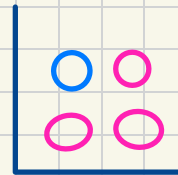
Example. Which Box!



Box #1



Box #2



Box #3

Curtain.

POSTERIOR PROBABILITIES

Q: $P[\text{Box \#}i \mid \text{Pink}] =$

$$= \frac{P[\text{Box \#}i \cap \text{Pink}]}{P[\text{Pink}]}$$

$$= \frac{P[\text{Box \#}i] \cdot P[\text{Pink} \mid \text{Box \#}i]}{\sum_{k=1}^3 P[\text{Box \#}k] \cdot P[\text{Pink} \mid \text{Box \#}k]}$$

Modeling Choice

$$P[\text{Box \#}i] = \frac{1}{3}, \forall i$$

$$P[\text{Box \#}i] = \frac{\text{\# of marbles in Box \#}i}{\text{total \# of marbles}}$$

PRIOR PROBAB.

$$\rightarrow P[Y = "k" \mid X = x] = ?$$